



Effects of dietary starch and fatty acid supplementation on milk production and metabolic responses during the immediate postpartum period in dairy cows

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ABSTRACT

This study aimed to investigate the interaction between dietary starch and fatty acid (FA) supplementation on production responses in multiparous Holstein cows. Sixty multiparous Holstein cows were used in a randomized complete block design and assigned to one of 4 treatments in a 2×2 factorial arrangement. The following treatment diets were fed from 1 to 23 d postpartum (fresh period; FR): (1) a diet containing 22% DM starch and no supplemental FA (LS); (2) a diet containing 28% DM starch and no supplemental FA (HS); (3) a diet containing 22% DM starch and a calcium salt containing 70% C16:0 and 20% *cis*-9 C18:1 (LS+FA); and (4) a diet containing 28% DM starch and a calcium salt containing 70% C16:0 and 20% *cis*-9 C18:1 (HS+FA). The starch concentration of the treatment diets was altered by partially replacing ground corn with soyhulls and soybean meal. The FA supplement was added at 2.6% of diet DM to deliver ~2.0% of FA and replaced soyhulls in the FA-supplemented diets. During the carryover period (CO; 24–44 d postpartum), all cows were offered a common diet (27.8% starch and 3.05% FA of DM) to evaluate the carryover effects of the treatment diets. During FR, dietary starch and FA supplementation interacted with time to affect DMI, plasma BHB, and tended to interact to affect ECM yield and milk lactose content. The starch $\times$ FA interaction affected the yields of milk, milk fat, milk lactose, 3.5% FCM, ECM, and tended to affect milk protein yield and FCM/DMI. Based on pairwise comparisons among treatment combinations, HS increased milk and lactose yields compared with the other treatments. The HS and LS+FA diets increased milk fat yield compared with LS, and HS increased 3.5% FCM and ECM yields compared with LS, whereas LS+FA tended to increase 3.5% FCM yield compared with LS. Overall, increasing dietary starch increased milk and

lactose yields, tended to increase DMI, and reduced plasma BHB concentration. Supplemental FA increased FCM/DMI and plasma nonesterified fatty acid and BHB concentrations; reduced DMI, BW, and plasma insulin and glucose concentrations; and tended to increase ECM/DMI and BCS loss. During the CO, increasing dietary starch during FR increased milk lactose yield and tended to increase milk yield and milk lactose content, whereas FA supplementation reduced BW, tended to reduce milk lactose content, and tended to increase BCS loss. In conclusion, the effect of FA supplementation on milk production of early-lactation cows depended on dietary starch level. Additionally, feeding high-starch diets during FR improved milk yield, with effects that were partially carried over into established lactation, indicating that early FP diet can influence performance in the weeks thereafter.

Key words: supplemental, starch, fatty acid, early lactation

INTRODUCTION

Dairy cows experience negative energy balance (NEB) during the immediate postpartum period due to reduced DMI and increased energy demands for milk production (Martens, 2023). As a result of NEB, cows compensate by mobilizing their body's energy reserves by increasing the flow of nonesterified fatty acids (NEFA) to the liver, peripheral tissues, and mammary gland for milk fat synthesis (Drackley, 1999). However, excessive adipose tissue mobilization may impair production and reproduction performance (Mann, 2022), elevate the risk of developing metabolic disorders (Contreras and Sordillo, 2011), and compromise immune function (Contreras et al., 2018). Consequently, strategies that increase dietary energy intake could help to mitigate the severity of NEB (Mekuriaw, 2023). Strategies used to increase dietary energy density and energy intake during the immediate postpartum period include increasing dietary starch and fatty acid (FA) supplementation (Albornoz and Allen, 2018;

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

de Souza et al., 2021). Studies such as these underscore the importance of nutritional interventions in supporting productive and healthy transitions for dairy cows.

The effect of increasing dietary starch on milk production of early-lactation cows has been inconsistent. Some studies have reported increased milk yield without affecting the yields of milk components (McCarthy et al., 2015), whereas others have found no effect (Piccioli-Cappelli et al., 2022). Importantly, the impact of dietary starch on production performance during the immediate postpartum period depends not only on the total starch content, but also on the rumen fermentability of the starch source, with highly fermentable sources reducing DMI and milk component yields compared with less fermentable sources (Albornoz and Allen, 2018). Furthermore, a high-starch diet (~30% DM) in post-peak dairy cows' diets can increase the risk of milk fat depression (MFD) and enhance energy deposition in adipose tissue, especially when dietary forage NDF (fNDF) is reduced (Boerman et al., 2015a). Overall, when formulating diets for early-lactation cows, both the amount and fermentability of dietary starch should be considered in order to optimize milk yield and composition and minimize risks such as MFD.

The traditional recommendation was not to increase dietary FA in early-lactation cows (Grummer, 1992) due to possible adverse effects on DMI and energy metabolism (Kuhla et al., 2016). However, recent studies have reported that FA supplementation has a positive effect on production responses, and the magnitude of this effect is related to the FA profile of the supplement (Lock et al., 2025). Supplementation with palmitic acid (C16:0) in the immediate postpartum period increased ECM yield by 3.9 kg/d without affecting DMI or BW loss (Parales-Girón et al., 2025) and by 5.3 kg/d with a tendency to increase BW loss (de Souza and Lock, 2019). This suggests that targeted FA supplementation can enhance productivity without compromising cow health during early lactation. Feeding FA blends with different C16:0 and oleic acid (*cis*-9 C18:1) ratios in the immediate postpartum period increased milk fat and ECM yields compared with a control diet without supplemental FA, with a linear reduction in BW loss associated with increasing *cis*-9 C18:1 content (de Souza et al., 2021). Furthermore, duodenal infusion of *cis*-9 C18:1 has limited lipolysis by promoting lipogenesis in adipose tissue of early lactation dairy cows (Abou-Rjeileh et al., 2023; 2025). Overall, recent evidence indicates that specific FA supplementation strategies can optimize production outcomes in early-lactation cows and minimize negative effects on energy balance.

Few studies have evaluated the combined effects of dietary starch content and FA supplementation on dairy

cow performance. Daneshvar et al. (2021) compared 2 levels of dietary starch and supplementation with either palmitic acid (C16:0) or stearic acid (C18:0) during the first 4 wk of lactation and reported that milk yield increased with the high-starch diet when supplemented with C18:0, whereas C16:0 tended to increase milk yield in low-starch diets. Piantoni et al. (2015) and Weiss and Pinos-Rodríguez (2009) evaluated different fNDF content and a saturated FA supplement in the immediate postpartum period and established lactation, respectively. As a result of fNDF content manipulation, the low-fNDF and high-fNDF diets resulted in high and low dietary starch levels, respectively. Piantoni et al. (2015) observed that FA supplementation promoted energy partitioning to body energy reserves at the expense of milk yield, particularly in a low-fNDF high-starch diet. In contrast, Weiss and Pinos-Rodríguez (2009) found that cows fed supplemental FA in a low-fNDF high-starch diet partitioned energy toward milk rather than body reserves. These contrasting findings suggest that the timing of FA supplementation and the lactation stage may influence whether energy is directed toward milk production or body reserves, highlighting the complexity of nutritional strategies in early-lactation dairy cows.

No studies have evaluated the effects of different dietary starch levels and FA supplements containing C16:0 and *cis*-9 C18:1 on milk production in early-lactation dairy cows while also maintaining similar dietary fNDF content. Therefore, our objective was to determine the effects of dietary starch and FA supplementation on milk production of early-lactation dairy cows. We hypothesized that increasing dietary starch in the immediate postpartum period would increase milk yield and reduce milk fat content. We further hypothesized that FA supplementation would increase milk fat yield and milk fat content, mitigating any reduction in milk fat with a higher dietary starch diet.

MATERIALS AND METHODS

Animal Housing and Care

All experimental procedures were approved by the Institutional Animal Care and Use Committee at Michigan State University (East Lansing, MI). The experiment began on September 15, 2021, and ended on April 8, 2022. Cows were fed once daily (0745 h) at 115% of expected intake during the fresh period (FR) and carryover period (CO) and milked 3 times daily (0400, 1200, and 2000 h). Water was available ad libitum in each stall, and stalls were bedded with sawdust and cleaned twice daily. Standard reproduction, herd health checks, and breeding practices were maintained throughout the study.

Design and Treatment Diets

Sixty multiparous Holstein cows at the Michigan State University Dairy Cattle Teaching and Research Center (East Lansing, MI) were used in a randomized complete block design experiment with a 2×2 factorial arrangement of treatments. Cows were blocked by previous lactation 305-d mature equivalent milk production (mean $\pm$ SD; $13,212 \pm 1,520$ kg), BCS (up to 0.50-unit difference using a scale of 1 = thin and 5 = fat in 0.25-unit increments, BCS 3.40 ± 0.15), and projected calving date (34 ± 18 d). The BCS used to block cows was the measurement recorded before parturition. Cows within each block were randomly assigned to one of 4 treatment diets on their enrollment at 21 d before their expected parturition date. The number of cows per treatment ($n = 15$ per treatment) provided $>80\%$ power to detect a 5% difference in 3.5% FCM yield, based on variability observed in prior studies evaluating 2 levels of NDF and FA in a 2×2 factorial design (Piantoni et al., 2015) and 2 levels of chromium and FA supplementation in a 2×2 factorial design (Parales-Girón et al., 2025).

The treatments were as follows: (1) a diet containing 22% DM starch and no supplemental FA (LS); (2) diet containing 28% DM starch and no supplemental FA (HS); (3) diet containing 22% DM starch and a calcium salt containing 70% C16:0 and 20% *cis*-9 C18:1 (LS+FA); (4) diet containing 28% DM starch and a 70% C16:0 and 20% *cis*-9 C18:1 calcium salt (HS+FA). The starch concentration of the treatment diets was adjusted by partially replacing ground corn with soyhulls and soybean meal. The FA supplement (Spectrum Complete, Perdue Agribusiness, Salisbury, MD) was added at 2.59% DM to deliver $\sim 2.00\%$ DM of FA and replaced soyhulls in the FA-supplemented diets. Dietary fNDF and CP contents were held constant across treatment diets. Treatment diets were offered from 1 to 23 d postpartum during the FR, and all cows were fed a common diet to evaluate possible carryover effects of the treatment diets from d 24 to 44 d postpartum (CO). Treatment diets were mixed daily in a tumble-mixer and fed from the morning following parturition, and common diets (close-up and carryover) were mixed daily in a mixer wagon. The DM concentration of forages was determined twice weekly using a Koster Moisture Test (Koster Moisture Testers Inc.), and diets were adjusted when necessary. The ingredient and nutrient composition of the diets fed as TMR, including the close-up and carryover rations for reference, are described in Table 1. All rations were formulated using the Cornell Net Carbohydrate and Protein System (CNCPS, v. 6.5.5; AMTS.Cattle.Professional v. 4.17.0.0, AMTS LLC; Van Amburgh et al., 2015). Feed access was blocked from 0545 to 0745 h for orts collection and new feed offer-

ings. The acronyms LS, HS, LS+FA, and HS+FA are used only in reference to the low-starch, high-starch, low-starch plus FA, and high-starch plus FA treatments, respectively.

Data and Sample Collection

Milk yield, feed offered, and feed refused were recorded daily throughout the experiment. All samples and body measurements were collected or recorded on the same days of the week. Therefore, there is a variation of ± 2 d relative to the average DIM reported for data and sample collection. Samples of all diet ingredients (0.5 kg) and orts from each cow ($\sim 12.5\%$) were collected weekly during the FR, and diet ingredients were collected monthly during the CO and stored at -20°C until being processed. Milk samples were collected twice a week at each milking during the FR and once a week during the CO and stored with preservative (Bronolab W-II liquid, Advanced Instruments, Norwood, MA) at 4°C for component analysis. During the FR, milk samples for component analysis were collected on d 6 and 9, 13 and 16, and 20 and 23 postpartum, with the averages from these sets of days representing wk 1, 2, and 3, respectively. During the CO, collection days for milk components were d 27, 34, and 41, representing wk 4, 5, and 6, respectively. An additional milk sample was collected at each milking on d 6, 13, and 20 postpartum and stored without preservative at -20°C for FA analysis.

Blood samples were collected weekly during the FR on d 9, 16, and 23 postpartum by puncture of coccygeal vessels 1 h before feeding. Blood was collected into 2 evacuated tubes containing potassium EDTA. Both tubes were centrifuged at $3,000 \times g$ for 15 min at 3°C immediately after sample collection, and plasma was harvested and stored at -20°C until analysis. Body weight was recorded weekly from -21 d of the expected parturition day until calving and 3 times per week after calving. Body condition score was recorded once 3 wk before the expected parturition date and then weekly after calving by 3 trained investigators on a 5-point scale, as described by Wildman et al. (1982).

Sample Analysis

Feed and orts samples were dried in a 55°C forced-air oven for 72 h, and DM content was determined. Before drying, feed ingredients were composited by month. Orts from individual cows were dried to calculate weekly DMI during the FR. Once dried, samples of feed ingredients were ground in a Wiley mill (1-mm screen; Arthur H Thomas Co., Philadelphia, PA) and analyzed for ash, CP, NDF, and starch as described by Boerman et al. (2017), and FA as described by Bales et al. (2024).

Table 1. Ingredient and nutrient composition of close-up, treatment, and carryover diets

Item	Diet					
	Close-up	Treatment ¹				Carryover
		LS	HS	LS+FA	HS+FA	
DM, %	49.5	44.9	44.1	45.1	44.9	44.6
Ingredient, % DM						
Corn silage	37.9	24.8	24.8	24.9	24.9	38.7
Alfalfa silage	—	23.3	23.2	23.4	23.7	13.4
Alfalfa hay	—	7.16	7.10	7.34	7.30	—
Grass hay	27.5	—	—	—	—	—
Ground corn	4.41	17.1	25.4	16.9	25.1	20.6
Soybean meal	5.86	5.43	6.17	5.83	6.46	6.76
Soyhulls	5.54	12.1	3.20	8.98	—	4.80
Whole cottonseed	—	2.22	2.22	2.20	2.20	6.84
DCAD supplement close-up ²	9.45	—	—	—	—	—
Protein supplement ³	—	6.07	6.14	6.00	6.00	7.03
Fatty acid supplement ⁴	—	—	—	2.59	2.59	—
Mineral, vitamin, and protein mix close-up ⁵	9.30	—	—	—	—	—
Mineral and vitamin mix ⁶	—	1.83	1.83	1.81	1.80	1.97
Nutrient composition, ⁷ % DM						
NDF	38.7	35.6	30.3	33.9	28.8	31.6
fNDF	33.1	23.4	23.3	23.8	23.8	20.0
CP	16.6	16.6	16.7	16.5	16.5	16.1
Starch	17.1	21.1	27.3	20.9	27.0	27.8
FA	1.69	2.22	2.44	4.19	4.39	3.05
16:0	0.31	0.38	0.40	1.78	1.80	0.54
18:0	0.05	0.06	0.06	0.20	0.20	0.08
<i>cis</i> -9 18:1	0.37	0.35	0.42	0.66	0.72	0.54
<i>cis</i> -9, <i>cis</i> -12 18:2	0.73	1.02	1.15	1.07	1.18	1.63
<i>cis</i> -9, <i>cis</i> -12, <i>cis</i> -15 18:3	0.14	0.30	0.30	0.29	0.28	0.20

¹LS = diet containing starch at 22% DM and no supplemental FA; HS = diet containing starch at 28% DM and no supplemental FA; LS+FA = diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; HS+FA = diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1.

²Soychlor (West Central Soy, Ralston, IA).

³Protein supplement contained 40% sodium sesquinate, 20% calcium carbonate, 19% ground corn, 15% blood blend (Spectrum AgriBlue, Perdue Agribusiness, Salisbury, MD), 3% rumen-protected Met (Smartamine M, Adisseo, Alpharetta, GA), and 2% urea.

⁴Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1 (Spectrum Complete, Perdue Agribusiness, Salisbury, MD). The supplement contained (g/100 g of fatty acids) 1.47 of C14:0, 70.8 of C16:0, 7.20 of C18:0, 16.0 of *cis*-9 C18:1, and 77.2% total fatty acids.

⁵Mineral, vitamin, and protein mix used in close-up diet contained (DM basis) 77.1% heat-treated soybean meal (Amino Plus, Ag Processing Inc., Mason City, IA), 8.47% magnesium sulfate salt, 7.80% blood blend (Spectrum Agribusiness, Salisbury, MD), 2.56% animal fat, 1.34% magnesium oxide, 1.88% white salt, and 0.85% trace minerals and vitamins.

⁶Mineral and vitamin mix contained 26% bypass protein (Amino Plus, Ag Processing Inc., Mason City, IA), 14% calcium carbonate, 13% blood meal (Caledonia Pass, Papillon), and sodium sesquicarbonate, 9% calcium phosphate, 6% urea, 5% white salt, 4% magnesium oxide, 3% calcium magnesium carbonate (MIN AD, Min Ad Inc., Winnemucca, NV) and potassium chloride red, 2% potassium electrolyte mix (Ion Plus, D&D Ingredients LLC, Delphos, OH), 1% selenium, and <1% ferrous sulfate, trace minerals, (Smartamine M, Adisseo, Alpharetta, GA), vitamin D, vitamin E, and vitamin A.

⁷Analysis of CP, NDF, and starch performed at Cumberland Valley Analytical Services Inc. (Waynesboro, PA) as described by Boerman et al. (2017) and FA analysis performed internally as described by Bales and Lock (2024).

Milk samples were analyzed for fat, true protein, and lactose concentrations by mid-infrared spectroscopy (AOAC, 1990; method 972.160) by the Michigan Dairy Herd Improvement Association (CentralStar Michigan Lab, Grand Ledge, MI). Yields of 3.5% FCM, ECM, and milk components were calculated using milk yield and component concentrations from each milking, summed for a daily total, and averaged for each week. Milk lipids were extracted, and FA-methyl esters prepared and quantified using gas chromatography, according to Bales

and Lock (2024). Yields of individual FA in milk fat (g/d) were calculated using milk fat yield and FA concentrations to determine yield on a mass basis using the molecular weight of each FA and correcting for glycerol content and other milk lipid classes (Benoit et al., 2024). Plasma samples were analyzed at the Michigan State University Veterinary Diagnostic Laboratory (Lansing, MI) for concentrations of NEFA, BHB, and glucose using an Olympus AU640e chemistry analyzer (Olympus America, Center Valley, PA), and plasma insulin was quantified

with a bovine insulin ELISA using a solid-phase 2-site enzyme immunoassay (Bovine Insulin ELISA, Mercodia AB, Uppsala, Sweden).

Statistical Analysis

Data were analyzed separately for FR and CO. All weekly data were analyzed using the GLIMMIX procedure of SAS v.9.4 (SAS Institute Inc. Cary, NC), with week of lactation being the repeated measure. The model is as follows:

$$Y_{ijklmn} = \mu + F_i + L_k + T_l + F_iL_k + F_iT_l + L_kT_l + F_iL_kT_l + B_m + C(B_mF_iL_k)_j + J_n + e_{ijklmn},$$

where μ = overall mean; F_i = fixed effect of dietary starch ($i = 1$ to 2); L_k = fixed effect of FA supplementation ($k = 1$ to 2); T_l = fixed effect of time ($l = 1$ to 3); F_iL_k = fixed effect of the interaction between starch and FA; F_iT_l = fixed effect of the interaction between starch and time; L_kT_l = fixed effect of the interaction between FA and time; $F_iL_kT_l$ = fixed effect of the interaction between starch, FA, and time; B_m = random effect of block ($m = 1$ to 15); $C(B_mF_iL_k)_j$ = random effect of cow within block and treatment diet; J_n = random effect of Julian date ($n = 1$ to 46); and e_{ijklmn} = residual error. A reduced model excluding the effects of time and the effect of cow within block and treatment was used to determine the effect of treatments on variables that were recorded once per period. First-order autoregressive was the covariate structure used for analysis because it resulted in the smallest Bayesian information criterion for most measured variables. Precalving BCS and BW were used as covariates for BW, BCS, BW change, and BCS change. Significance was declared at $P \leq 0.05$ for main effects and at $P \leq 0.10$ for interactions, whereas tendencies were declared at $0.05 < P \leq 0.10$ for main effects and at $0.10 < P \leq 0.15$ for interactions. When a 3-way interaction (starch $\times$ FA $\times$ time) was $P \leq 0.15$, the SLICE option in the LSMEANS statement in SAS was used to evaluate treatment effects within week of lactation. When a 2-way starch $\times$ FA interaction was $P \leq 0.15$, all pairwise comparisons among the 4 treatment combinations were obtained using the PDIF option of the LSMEANS statement (protected least significant difference test) following a similar approach to the one used in dos Santos Neto et al. (2023). Data are presented as LSM and their associated SEM. All cows were in apparent good health at the beginning of the study, and treatment groups were not different in terms of 305-d mature equivalent yield ($13,212 \pm 1,520$ kg; $P = 0.77$), BW (782 ± 70 kg; $P = 0.37$), or BCS (3.40 ± 0.15 ; $P = 0.87$) before calving.

RESULTS

Health Events

A summary of health events is presented in Supplemental Table S1 (see Notes). During the FR, 9 health events were recorded; milk fever was the primary health event observed, followed by ketosis. The number of health events was lower during the CO, with 1 mastitis event.

Production Responses: FR

Dietary starch level and FA supplementation interacted with time to affect DMI ($P = 0.05$; Figure 1A) and tended to interact to affect ECM yield ($P = 0.12$; Figure 1C) and lactose content ($P = 0.14$; Supplemental Table S2, see Notes). Starch and FA supplementation interacted to modify the yields of milk ($P = 0.07$; Figure 1B), milk fat ($P = 0.06$; Figure 2A), milk lactose ($P = 0.06$; Table 2), 3.5% FCM ($P = 0.04$; Figure 2B), and ECM ($P = 0.04$; Figure 1C), and tended to interact to modify milk protein yield ($P = 0.15$; Figure 2C) and 3.5% FCM/DMI ($P = 0.15$; Table 2). The HS treatment increased the yields of milk and milk lactose compared with LS, LS+FA, and HS+FA ($P \leq 0.02$). The HS and LS+FA treatments increased milk fat yield compared with LS ($P = 0.05$ and $P = 0.04$), and HS+FA had intermediate values. The HS treatment increased the yields of 3.5% FCM and ECM compared with LS ($P = 0.02$), and LS+FA tended to increase 3.5% FCM compared with LS ($P = 0.10$). The HS treatment increased milk protein yield compared with HS+FA ($P = 0.03$), and the LS+FA treatment increased 3.5% FCM/DMI compared with LS ($P = 0.02$). We did not observe any interactions between dietary starch and FA supplementation for milk lactose content, BW, BCS, and changes in BW and BCS ($P \geq 0.29$; Table 2).

Overall, increasing dietary starch increased milk yield by 2.29 kg/d ($P = 0.05$), lactose yield by 0.13 kg/d ($P = 0.03$), and milk lactose content by 0.05 g/100 g ($P = 0.05$); tended to increase DMI by 0.90 kg/d ($P = 0.10$); and tended to reduce milk protein content by 0.11 g/100 g ($P = 0.06$; Table 2). Overall, FA supplementation increased milk fat content by 0.25 g/100 g ($P < 0.01$), FCM/DMI by 0.14 kg/kg ($P = 0.05$); reduced DMI by 1.09 kg/d ($P = 0.05$), BW by 14 kg ($P = 0.03$), BCS by 0.07 units ($P = 0.03$); tended to increase ECM/DMI by 0.11 kg/kg ($P = 0.08$) and BCS loss by 0.03 units/wk ($P = 0.06$); and tended to reduce milk lactose content by 0.04 g/100 g ($P = 0.10$; Table 2).

Protein supply predictions from the CNCPS model (v. 6.5.5) using dietary ingredients (Table 1) and animal performance (Table 2) for the experimental diets are reported in Supplemental Table S3 (see Notes). We did

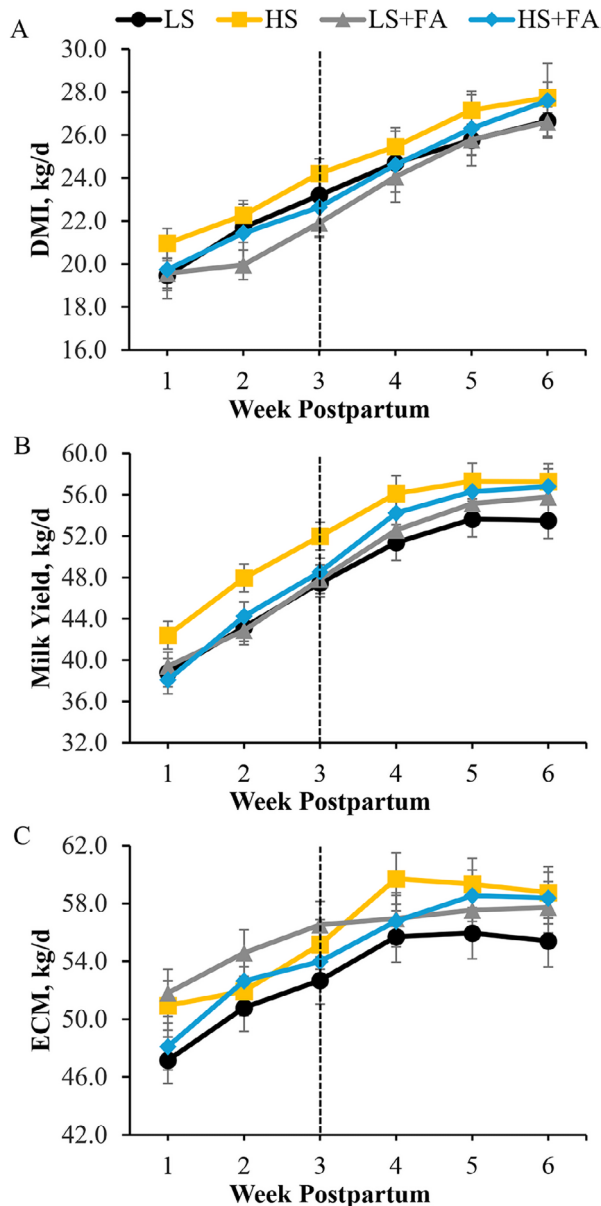


Figure 1. Effects of dietary treatments on DMI (A), milk yield (B), and ECM yield (C) during the FR (1–23 d postpartum) and CO (24–44 d postpartum). Diets fed during the FR were LS (diet containing starch at 22% DM and no supplemental FA; black line), HS (diet containing starch at 28% DM and no supplemental FA; yellow line), LS+FA (diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; gray line), and HS+FA (diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; blue line). All experimental weeks are ± 2 DIM. The vertical dashed line at wk 3 indicates the start of the CO when all cows were fed a common diet containing starch at 28% DM with no supplemental FA. Dietary starch and FA supplementation interacted with time to affect DMI ($P = 0.05$). During the CO, we did not observe a treatment effect on DMI ($P \geq 0.12$). During the FR, dietary starch and FA supplementation interacted to affect milk yield ($P = 0.07$). During CO, a higher dietary starch tended to increase milk yield compared with lower dietary starch ($P = 0.09$). During the FR, dietary starch and FA interacted to influence ECM yield ($P = 0.04$). During the CO, we did not observe interactions between dietary starch and FA supplementation and no treatment effects on ECM yield ($P \geq 0.19$). Error bars indicate SEM.

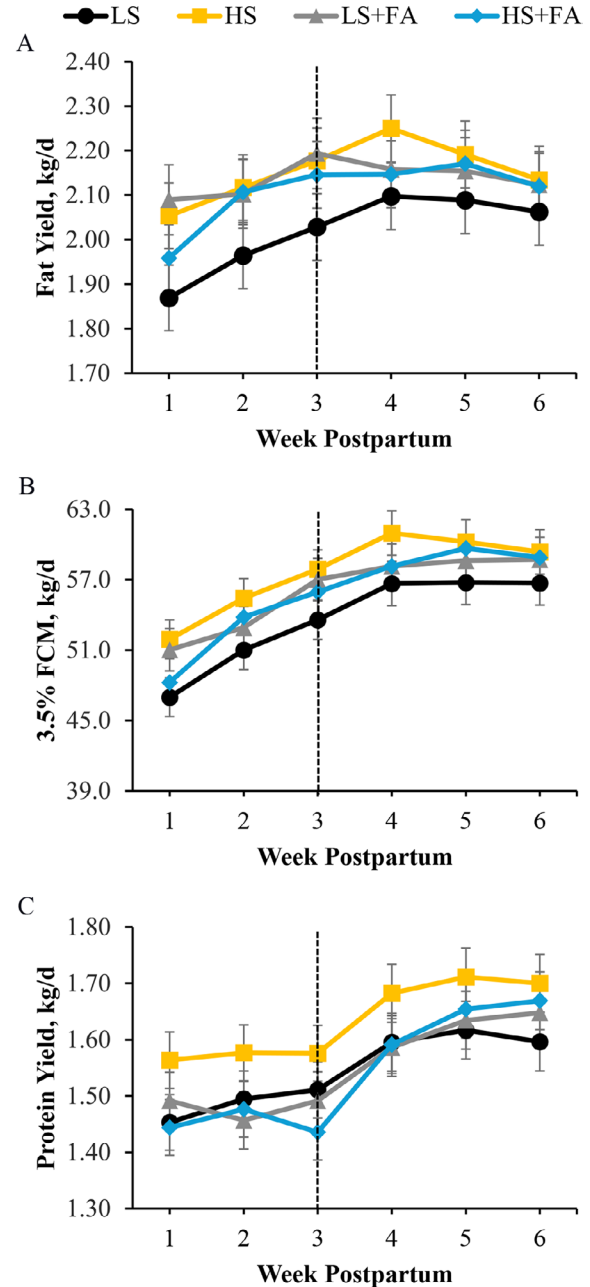


Figure 2. Effects of dietary treatments on milk fat yield (A), 3.5% FCM yield (B), and milk protein yield (C) during the FR (1–23 d postpartum) and CO (24–44 d postpartum). Diets fed during the FR were LS (diet containing starch at 22% DM and no supplemental FA; black line), HS (diet containing starch at 28% DM and no supplemental FA; yellow line), LS+FA (diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; gray line), and HS+FA (diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; blue line). All experimental weeks are ± 2 DIM. The vertical dashed line at wk 3 indicates the start of the CO when all cows were fed a common diet containing starch at 28% DM with no supplemental FA. During the FR, dietary starch and FA supplementation interacted to affect the yields of milk fat ($P = 0.06$) and 3.5% FCM ($P = 0.04$) and tended to interact to influence milk protein yield ($P = 0.15$). During the CO, we did not observe interactions between dietary starch, FA supplementation, and time, nor were there treatment effects on the yields of milk fat, milk protein, or 3.5% FCM ($P \geq 0.34$). Error bars indicate SEM.

Table 2. Dry matter intake, milk production, milk composition, BW, and BCS for cows fed treatment diets during the FR (d 1–23 postpartum)

Variable	Treatment ¹					P-value ²						
	LS	HS	LS+FA	HS+FA	SEM	Starch	FA	Starch × FA	time	Starch × time	FA × time	Starch × FA × time
DMI, kg/d	21.5	22.5	20.5	21.3	0.63	0.10	0.05	0.83	<0.01	0.90	0.25	0.05
Yield, kg/d	43.1 ^b	47.4 ^a	43.3 ^b	43.6 ^b	1.30	0.05	0.11	0.07	<0.01	0.01	0.78	0.44
Milk	1.95 ^b	2.12 ^a	2.13 ^a	2.07 ^{ab}	0.07	0.36	0.26	0.06	<0.01	0.59	0.99	0.30
Fat	1.49	1.57	1.48	1.45	0.05	0.46	0.11	0.15	0.67	0.30	0.58	0.23
Protein	2.11 ^b	2.35 ^a	2.14 ^b	2.16 ^b	0.06	0.03	0.13	0.06	<0.01	0.44	0.78	0.39
Lactose	50.5 ^b	55.1 ^a	53.7 ^{ab}	52.7 ^{ab}	1.51	0.17	0.78	0.04	<0.01	0.31	0.88	0.17
3.5% FCM ³	50.2 ^b	54.3 ^a	52.7 ^{ab}	51.6 ^{ab}	1.48	0.24	0.92	0.04	<0.01	0.40	0.89	0.12
ECM ⁴												
Content, g/100 g												
Fat	4.57	4.48	4.83	4.71	0.10	0.25	<0.01	0.87	<0.01	0.48	0.13	0.73
Protein	3.46	3.30	3.38	3.33	0.06	0.06	0.66	0.34	<0.01	0.48	0.03	0.48
Lactose	4.85	4.90	4.80	4.86	0.03	0.05	0.10	0.98	<0.01	0.81	0.84	0.14
FCM/DMI, kg/kg	2.35	2.46	2.59	2.50	0.07	0.87	0.05	0.15	0.06	0.25	0.08	0.70
ECM/DMI, kg/kg	2.34	2.44	2.54	2.46	0.07	0.87	0.08	0.17	0.01	0.35	0.12	0.62
BW, kg	721	722	705	711	6.77	0.56	0.03	0.67	<0.01	0.94	0.12	0.91
BW change, kg	-2.43	-2.57	-2.49	-2.40	0.32	0.94	0.84	0.69	—	—	—	—
BCS	3.24	3.23	3.16	3.18	0.03	0.82	0.03	0.56	<0.01	0.41	0.35	0.85
BCS change	-0.05	-0.09	-0.10	-0.11	0.02	0.16	0.06	0.29	—	—	—	—

^{a,b}Means within a row with different superscripts differ ($P \leq 0.05$). Superscripts are shown only when a starch × FA interaction was $P \leq 0.15$. Pairwise comparisons among treatment combinations were obtained using the PDIF option of the LSMEANS statement in SAS. When a three-way interaction (starch × FA × time) was $P \leq 0.15$, treatment effects were compared within week of lactation.

¹LS = diet containing starch at 22% DM and no supplemental FA; HS = diet containing starch at 28% DM and no supplemental FA; LS+FA = diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; HS+FA = diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1.

²P-values refer to the ANOVA results for the main effect of starch, the main effect of FA, the main effect of time, the interaction between starch and FA, the interaction between starch and time, the interaction between FA and time, and the interaction between starch, FA, and time. A dash (—) indicates that the effects do not apply to those variables.

³3.5% FCM = [(0.4324 × kg of milk) + (16.216 × kg of milk fat)] (NRC, 2001).

⁴ECM = (0.327 × kg of milk) + (12.95 × kg of milk fat) + (7.20 × kg of milk protein). This equation corrects milk to a 0.68 Mcal/kg energy basis.

Table 3. Summation and individual milk fatty acid (FA) yield for cows fed treatment diets during the FR (d 1–23 postpartum)

Variable	Treatment ¹					P-value ²						
	LS	HS	LS+FA	HS+FA	SEM	Starch	FA	Starch × FA	Time	Starch × time	FA × time	Starch x FA × time
Summation by source, ³ g/d												
De novo	396	427	367	376	17.9	0.18	0.01	0.46	<0.01	0.99	0.08	0.64
Mixed	592	625	700	709	21.9	0.26	<0.01	0.54	<0.01	0.72	0.75	0.31
Preformed	875	935	926	913	37.8	0.49	0.67	0.29	0.11	0.30	0.17	0.93
Selected individual FA, g/d												
C4:0	65.8	69.2	72.8	71.9	2.72	0.60	0.04	0.36	0.42	0.44	0.02	0.31
C6:0	38.7	41.6	37.9	38.9	1.82	0.18	0.25	0.52	<0.01	0.89	0.76	0.91
C8:0	20.9	22.9	19.1	19.7	1.19	0.19	0.01	0.45	<0.01	0.98	0.18	0.90
C10:0	45.8	50.1	38.7	40.5	3.03	0.24	<0.01	0.62	<0.01	0.92	0.01	0.74
C12:0	48.4	52.6	40.3	42.0	3.08	0.26	<0.01	0.63	<0.01	0.82	<0.01	0.57
C14:0	166	179	148	153	7.31	0.14	<0.01	0.52	<0.01	0.99	<0.01	0.40
C16:0	550	579	658	669	20.2	0.26	<0.01	0.63	<0.01	0.79	0.84	0.26
cis-9 C16:1	40.7 ^{ab}	46.4 ^a	41.9 ^{ab}	40.2 ^b	2.35	0.36	0.26	0.09	<0.01	0.68	0.03	0.33
C18:0	236	247	244	248	10.4	0.44	0.65	0.75	<0.01	0.61	0.17	0.62
trans-6 to 8 C18:1	4.02	4.45	4.78	4.83	0.19	0.20	<0.01	0.32	<0.01	0.12	0.21	0.52
trans-9 C18:1	2.89	3.17	3.27	3.41	0.13	0.08	0.01	0.58	<0.01	0.24	0.01	0.67
trans-10 C18:1	4.62	5.55	4.96	5.81	0.38	0.02	0.41	0.92	<0.01	0.21	0.25	0.92
trans-11 C18:1	15.1	15.8	16.2	17.2	0.82	0.30	0.11	0.81	0.27	0.16	0.23	0.74
cis-9 C18:1	452 ^b	496 ^{ab}	508 ^a	482 ^{ab}	23.3	0.68	0.31	0.09	<0.01	0.46	0.01	0.34
cis-11 C18:1	21.2	22.6	20.5	20.2	1.06	0.54	0.11	0.39	<0.01	0.31	0.28	0.73
cis-9,cis-12 C18:2	44.4	48.4	45.5	46.5	2.02	0.20	0.84	0.42	0.30	0.62	<0.01	0.13
cis-9,cis-12,cis-15 C18:3	9.39	10.1	9.34	8.99	0.44	0.62	0.14	0.18	0.06	0.26	0.75	0.08

^{a,b}Means within a row with different superscripts differ ($P \leq 0.05$). Superscripts are shown only when a starch × FA interaction was $P \leq 0.15$. Pairwise comparisons among treatment combinations were obtained using the PDIF option of the LSMEANS statement in SAS. When a three-way interaction (starch × FA × time) was $P \leq 0.15$, treatment effects were compared within week of lactation.

¹LS = diet containing starch at 22% DM and no supplemental FA; HS = diet containing starch at 28% DM and no supplemental FA; LS+FA = diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; HS+FA = diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1.

²P-values refer to the ANOVA results for the main effect of starch, the main effect of FA, the main effect of time, the interaction between starch and FA, the interaction between starch and time, the interaction between FA and time, and the interaction between starch, FA, and time.

³De novo FA originate from mammary de novo synthesis (<16 carbons), preformed FA originate from extraction from plasma (>16 carbons), and mixed FA originate from both sources (C16:0 plus *cis*-9 C16:1).

Table 4. Summation and individual milk fatty acid (FA) content for cows fed treatment diets during the FR (d 1–23 postpartum)

Variable	Treatment ¹					P-value ²						
	LS	HS	LS+FA	HS+FA	SEM	Starch	FA	Starch × FA	Time	Starch × time	FA × time	Starch × FA × time
Summation by source, ³ g/100 g of FA												
De novo	21.3	21.4	18.2	19.0	0.66	0.39	<0.01	0.47	0.02	0.44	<0.01	0.73
Mixed	31.9	31.5	35.2	35.5	0.39	0.89	<0.01	0.32	0.01	0.29	0.30	0.08
Preformed	46.6	47.1	46.5	45.6	0.88	0.75	0.29	0.37	<0.01	0.26	<0.01	0.29
Selected individual FA, g/100 g of FA												
C4:0	3.50	3.47	3.66	3.64	0.06	0.64	<0.01	0.84	<0.01	0.55	0.17	0.57
C6:0	2.08	2.11	1.91	1.95	0.05	0.39	<0.01	0.85	0.11	0.27	0.64	0.83
C8:0	1.12	1.15	0.94	0.99	0.05	0.34	<0.01	0.75	0.01	0.60	0.02	0.98
C10:0	2.49	2.51	1.90	2.06	0.13	0.43	<0.01	0.58	<0.01	0.30	<0.01	0.78
C12:0	2.64	2.64	1.99	2.13	0.14	0.56	<0.01	0.56	<0.01	0.22	<0.01	0.58
C14:0	8.98	8.98	7.36	7.73	0.30	0.46	<0.01	0.47	0.03	0.52	<0.01	0.44
C16:0	29.7	29.2	33.1	33.5	0.41	0.87	<0.01	0.23	0.14	0.22	0.20	0.06
cis-9 C16:1	2.20	2.30	2.09	2.01	0.07	0.89	<0.01	0.18	<0.01	0.09	<0.01	0.15
C18:0	12.6	12.4	12.3	12.4	0.30	0.90	0.51	0.46	<0.01	0.47	0.19	0.20
trans-6 to 8 C18:1	0.22	0.23	0.24	0.24	0.01	0.44	0.01	0.72	<0.01	0.03	0.06	0.92
trans-9 C18:1	0.15	0.16	0.16	0.17	0.004	0.10	0.01	0.73	<0.01	0.35	<0.01	0.48
trans-10 C18:1	0.25	0.28	0.25	0.29	0.02	0.03	0.69	0.72	0.43	0.10	0.33	0.83
trans-11 C18:1	0.82	0.79	0.82	0.85	0.03	0.85	0.28	0.32	0.01	0.14	0.02	0.51
cis-9 C18:1	24.4	24.9	25.3	24.1	0.69	0.50	0.91	0.12	0.09	0.35	<0.01	0.46
cis-11 C18:1	1.13	1.14	1.03	1.01	0.04	0.99	<0.01	0.74	0.03	0.55	0.17	0.85
cis-9,cis-12 C18:2	2.34	2.42	2.27	2.34	0.05	0.12	0.17	0.88	0.01	0.04	0.18	0.42
cis-9,cis-12,cis-15 C18:3	0.51	0.51	0.47	0.45	0.01	0.58	<0.01	0.41	0.03	0.19	0.44	0.42

¹LS = diet containing starch at 22% DM and no supplemental FA; HS = diet containing starch at 28% DM and no supplemental FA; LS+FA = diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% cis-9 C18:1; HS+FA = diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% cis-9 C18:1.

²P-values refer to the ANOVA results for the main effect of starch, the main effect of FA, the main effect of time, the interaction between starch and FA, the interaction between starch and time, the interaction between FA and time, and the interaction between starch, FA, and time.

³De novo FA originate from mammary de novo synthesis (<16 carbons), preformed FA originate from extraction from plasma (>16 carbons), and mixed FA originate from both sources (C16:0 plus cis-9 C16:1).

not estimate energy balance for treatments because no digestibility measurements were conducted (see Discussion).

Milk FA Content and Yields: FR

No interactions were observed between starch and FA supplementation for the yield of milk FA sources ($P \geq 0.29$) and most individual FA yields ($P \geq 0.18$), except for the yield of *cis*-9 C16:1 and *cis*-9 C18:1 ($P = 0.09$; Table 3). The HS treatment increased *cis*-9 C16:1 yield compared with HS+FA ($P = 0.05$), and LS+FA tended to increase *cis*-9 C18:1 compared with LS ($P = 0.06$; Table 3). Overall, increasing dietary starch increased the yield of *trans*-10 C18:1 by 0.89 g/d ($P = 0.02$) and did not affect the yields of the de novo, mixed, and preformed FA ($P \geq 0.18$; Table 3). Fatty acid supplementation increased the yield of C4:0 by 4.9 and C16:0 by 99 g/d ($P = 0.04$ and $P < 0.01$) and mixed FA by 97 g/d ($P < 0.01$), reduced de novo FA yields by 41 g/d ($P = 0.01$), and did not affect preformed FA yield ($P = 0.67$; Table 3).

Dietary starch and FA supplementation interacted with time to influence mixed milk FA content ($P = 0.08$) and C16:0 content ($P = 0.06$; Table 4). We did not observe interactions between dietary starch and FA for the content of milk FA sources ($P \geq 0.37$) or most individual milk FA ($P \geq 0.18$), except for a tendency to interact to affect *cis*-9 C18:1 content ($P = 0.12$; Table 4). However, no differences were found between FA-supplemented and nonsupplemented diets when compared within dietary starch for *cis*-9 C18:1 content ($P \geq 0.11$). Overall, dietary starch did not affect the content of milk FA sources ($P \geq 0.39$). Fatty acid supplementation increased the content of mixed FA and reduced de novo FA (both $P < 0.01$; Table 4).

Plasma Insulin and Metabolites: FR

Dietary starch and FA supplementation interacted with time to affect plasma BHB ($P = 0.07$; Supplemental Table S3). Overall, increasing dietary starch reduced plasma BHB by 2.26 mg/dL ($P = 0.02$; Table 5). Overall, FA supplementation increased plasma NEFA by 0.14 mmol/L and BHB by 2.12 mg/dL ($P \leq 0.03$) and reduced plasma insulin by 1.00 μ IU and glucose by 2.44 mg/dL ($P \leq 0.01$; Table 5).

Production Responses: CO

For milk production variables, we did not observe interactions between starch, FA, and time, nor between starch and FA ($P \geq 0.17$). Overall, increasing dietary starch increased milk lactose yield by 0.16 kg/d ($P = 0.05$) and tended to increase milk yield by 2.60 kg/d and

Table 5. Blood variables for cows fed treatment diets during the FR (d 1–23 postpartum)

Variable	Treatment ¹					P-value ²						
	LS	HS	LS+FA	HS+FA	SEM	Starch	FA	Starch × FA	Time	Starch × time	FA × time	starch × FA × time
Insulin, μIU	4.51	3.88	3.28	3.09	0.29	0.15	<0.01	0.44	0.08	0.42	0.83	0.28
Glucose, mg/dL	57.1	57.7	53.8	56.1	0.94	0.14	0.01	0.34	<0.01	0.61	0.23	0.90
NEFA, mmol/L	0.52	0.52	0.66	0.66	0.03	0.96	<0.01	0.95	0.04	0.91	0.92	0.25
BHB, mg/dL	9.74	7.76	12.1	9.60	1.00	0.02	0.03	0.77	0.70	0.16	<0.01	0.07

¹LS = diet containing starch at 22% DM and no supplemental FA; HS = diet containing starch at 28% DM and no supplemental FA; LS+FA = diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; HS+FA = diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1.

²P-values refer to the ANOVA results for the main effect of starch, the main effect of FA, the main effect of time, the interaction between starch and FA, the interaction between starch and time, the interaction between FA and time, and the interaction between starch, FA, and time.

milk lactose content by 0.03 g/100 g ($P \leq 0.10$). Overall, FA supplementation reduced BW by 14 kg ($P = 0.04$), tended to reduce milk lactose content by 0.03 g/100 g and BCS by 0.07 units ($P \leq 0.07$), and tended to increase BCS loss by 0.03 units/d ($P = 0.06$; Table 6).

DISCUSSION

In the immediate postpartum period, the gap between energy requirements for milk production and energy intake leads to NEB and mobilization of body reserves. Severe NEB increases the risk of metabolic diseases, reduced production, and impaired reproductive performance, so increasing dietary energy density is a common strategy to support early-lactation cows. Although high-producing post-peak cows may benefit more from high-starch diets (~30% DM; Boerman et al., 2015a), research indicates that increasing dietary starch during the immediate postpartum period using highly rumen-fermentable sources could negatively affect DMI and milk production (Albornoz and Allen, 2018). Regarding FA supplementation, our previous research has shown that feeding C16:0 during the immediate postpartum period increases milk production, with variable effects on BW loss (de Souza and Lock, 2019; Parales-Girón et al., 2025), whereas increasing *cis*-9 C18:1 in FA blends has been shown to both increase milk production and attenuate BW loss (de Souza et al., 2021). However, to our knowledge, no studies have evaluated the effect of dietary starch and a FA supplement containing 70% C16:0 and 20% *cis*-9 C18:1 in the immediate postpartum period. Therefore, the present study was designed to evaluate the possible interaction between these 2 factors on milk production responses in early-lactation dairy cows. Results provide evidence that cows fed a low-starch diet benefited from FA supplementation, resulting in increased milk fat and 3.5% FCM yields. In contrast, when cows were fed a high-starch diet, FA supplementation did not further improve milk production and was instead associated with a reduction in the yields of milk and lactose. Additionally, we did not observe interactions between dietary starch and FA supplementation or treatment effects on changes in BW and BCS. These findings suggest that adjusting FA supplementation based on dietary starch levels could enhance milk production in early-lactation cows, underscoring the importance of considering both energy source and FA supplement composition in dairy management, as supported by our results and previous research.

Studies evaluating different starch levels in postpartum dairy cows have yielded variable results. Increasing dietary starch by replacing forage with grain has generally been associated with greater milk production and DMI (Andersen et al., 2003; Rabelo et al., 2003) and replacing citrus pulp with ground corn similarly increased

milk yield and DMI (McCarthy et al., 2015). However, the fermentability of the starch source can strongly influence production outcomes. Highly fermentable starch sources, such as high-moisture corn, have been shown to reduce DMI and milk production compared with less fermentable sources such as ground corn (Albornoz and Allen, 2018). Research evaluating the combined effect of dietary starch and FA supplementation in dairy cows is limited. High-starch diets supplemented with C18:0 or low-starch diets plus C16:0 potentially increase energy partitioning to milk (Daneshvar et al., 2021), whereas a highly saturated FA supplement (46% C18:0 and 37% C16:0) in high-starch, low-forage NDF diets has been reported to either promote energy partitioning to body reserves or milk yield in the immediate postpartum period and established lactation, respectively (Piantoni et al., 2015; Weiss and Pinos-Rodríguez, 2009). These findings suggest that careful selection of starch sources and FA supplements is crucial for optimizing milk yield and cow health during the postpartum period.

In their review, Kuhla et al. (2016) suggested that supplemental FA during the immediate postpartum period may decrease DMI, potentially due to hepatic fuel oxidation influencing intake (Allen and Piantoni, 2013). Earlier research reported a limited impact on milk production with the use of UFA such as heat-treated soybean meal and tallow at dietary levels of 5% to 6% of DM, which may be related to total dietary FA concentration (5%–6% of DM), FA unsaturation level, and timing of supplementation. In contrast, more recent studies have demonstrated that supplementing FA blends rich in C16:0 and *cis*-9 C18:1 at 1.5% of DM in the immediate postpartum period positively affects milk production without compromising DMI or body energy reserves (de Souza et al., 2021), and supplemental C16:0 at 1.5% of DM in the immediate postpartum period increased milk production with variable effects on BW loss (de Souza and Lock, 2019; Parales-Girón et al., 2025). These findings suggest that targeted supplementation of specific FA during the immediate postpartum period may enhance milk production without negative effects on intake or energy reserves.

Contrary to our initial hypothesis, combining high starch with the FA supplement did not yield additive benefits to milk production. This absence of a synergistic effect on milk production could be attributed to several mechanisms. The supply of nutrients to the mammary gland is a function of mammary blood flow and the concentration of nutrients in the bloodstream, which in turn influence milk production in dairy cows (Davis and Collier, 1985). Therefore, reductions in DMI or nutrient digestibility can impair lactation performance. Although increasing dietary starch through partial forage substitution is typically associated with increased

Table 6. Dry matter intake, milk production, milk composition, BW, and BCS for cows fed treatment diets during the CO (d 24–44 postpartum)

Variable	Treatment ¹					P-value ²						
	LS	HS	LS+FA	HS+FA	SEM	Starch	FA	Starch × FA	Time	Starch × time	FA × time	Starch × FA × time
DMI, kg/d	25.7	26.8	25.5	26.2	0.67	0.12	0.45	0.72	<0.01	0.71	0.46	0.47
Yield, kg/d												
Milk	52.8	56.9	54.5	55.8	1.71	0.09	0.86	0.37	<0.01	0.22	0.13	0.78
Fat	2.08	2.19	2.15	2.15	0.07	0.42	0.91	0.42	0.05	0.71	0.50	0.55
Protein	1.60	1.70	1.62	1.64	0.05	0.19	0.63	0.34	<0.01	0.84	0.11	0.98
Lactose	2.65	2.86	2.69	2.80	0.09	0.05	0.87	0.54	<0.01	0.56	0.05	0.78
3.5% FCM ³	56.7	60.2	58.5	58.9	1.81	0.25	0.88	0.36	0.56	0.63	0.28	0.55
ECM ⁴	55.7	59.3	57.4	57.9	1.70	0.19	0.91	0.32	0.40	0.93	0.16	0.55
Content, g/100 g												
Fat	3.96	3.83	3.90	3.84	0.09	0.27	0.74	0.66	<0.01	0.09	0.51	0.22
Protein	3.04	2.97	2.96	2.93	0.05	0.34	0.20	0.70	0.21	0.38	0.58	0.47
Lactose	4.98	4.99	4.91	4.98	0.03	0.10	0.06	0.22	0.11	0.10	0.70	0.27
FCM/DMI, kg/d	2.21	2.26	2.30	2.26	0.05	0.95	0.37	0.34	<0.01	0.42	0.84	0.17
ECM/DMI, kg/d	2.19	2.23	2.26	2.22	0.05	0.97	0.48	0.37	<0.01	0.77	0.95	0.28
BW, kg	699	702	682	692	8.0	0.28	0.04	0.57	0.03	0.98	0.12	0.34
BW change, kg	-0.26	-0.34	-0.28	-0.20	0.17	0.99	0.70	0.63	—	—	—	—
BCS	3.08	3.05	2.99	3.00	0.04	0.85	0.07	0.63	<0.01	0.08	0.94	0.21
BCS change	-0.05	-0.09	-0.10	-0.11	0.02	0.16	0.06	0.29	—	—	—	—

¹LS = diet containing starch at 22% DM and no supplemental FA; HS = diet containing starch at 28% DM and no supplemental FA; LS+FA = diet containing starch at 22% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1; HS+FA = diet containing starch at 28% DM and a Ca salt containing 70% C16:0 and 20% *cis*-9 C18:1.

²P-values refer to the ANOVA results for the main effect of starch, the main effect of FA, the main effect of time, the interaction between starch and FA, the interaction between starch and time, the interaction between FA and time, and the interaction between starch, FA, and time. A dash (—) indicates that the effects do not apply to those variables.

³3.5% FCM = [(0.4324 × kg of milk) + (16.216 × kg of milk fat)] (NRC, 2001).

⁴ECM = (0.327 × kg of milk) + (12.95 × kg of milk fat) + (7.20 × kg of milk protein). This equation corrects milk to a 0.68 Mcal/kg energy basis.

DMI (Andersen et al., 2003; Rabelo et al., 2003), in our study, we observed increased DMI with increasing dietary starch while maintaining fNDF content. However, the addition of supplemental FA at 2% of dietary DM attenuated the increase in DMI. This was likely due to the increased flow of FA to the duodenum triggering the secretion gut peptides (e.g., cholecystokinin and peptide YY) that act as satiety signals by slowing gastric emptying and directly suppressing appetite centers in the brain, ultimately reducing feed intake (Bradford et al., 2008), and thus milk yield. Although FA supplementation at 1.5% of DM has shown benefits in milk production without compromising intake when dietary starch did not exceed 25% DM (de Souza and Lock, 2019; de Souza et al., 2021; Parales-Girón et al., 2025), inclusion above 2.0% DM has been associated with a reduction in DMI in post-peak cows (Rico et al., 2017). Additionally, high-starch diets can lower rumen pH (Salfer et al., 2018), potentially increasing the dissociation of Ca salts in the rumen (Sukhija and Palmquist, 1990). This would expose more UFA to rumen microbes. Unsaturated FA such as the ~20% *cis*-9 C18:1 in our supplement can disrupt cell membranes of fiber-digesting bacteria, impairing their growth (Maia et al., 2007; Sears et al., 2024). If fiber digestibility was indeed reduced, acetate production for *de novo* milk fat synthesis might have been limited (Urrutia et al., 2019). These interconnected mechanisms highlight the importance of balancing energy sources and FA supplementation to optimize nutrient utilization and milk yield in early-lactation dairy cows.

We also observed that increasing dietary starch led to higher yields of *trans*-10 C18:1, a FA commonly used as an indicator of shifts in rumen biohydrogenation pathways (Vahmani et al., 2025). Elevated production of *trans*-10 C18:1 in the rumen has been closely linked to decreased milk fat synthesis (Lock et al., 2007; Bauman et al., 2011). Importantly, we did not detect MFD, as milk fat yield and content in high-starch diets were not lower than those observed in low-starch diets. Interestingly, *cis*-9 C18:1 yield tended to be higher with LS+FA compared with LS, suggesting that oleic incorporation was favored under lower-starch conditions, likely due to a more stable rumen environment. *Cis*-9 C18:1 may have complemented palmitic acid to support milk fat synthesis through enhanced postruminal absorption or direct incorporation into milk fat.

Although no single factor or interaction can fully explain the lack of effect observed when supplemental FA were added to the high-starch diet, this outcome likely resulted from 3 concurrent mechanisms. First, reduced DMI and potential reduction in fiber digestibility with FA supplementation may have limited nutrient flow to the mammary gland, thereby constraining milk synthesis. Second, greater *trans*-10 C18:1 production under

high-starch conditions indicates altered rumen biohydrogenation that could have partially suppressed milk fat synthesis, even without detectable MFD. Third, the starch-dependent *cis*-9 C18:1 response suggests that oleic acid incorporation into milk fat was more efficient under low-starch, rumen-stable conditions. These mechanisms, when considered together may collectively help explain the absence of a synergistic effect on milk production when supplemental FA were included in the high-starch diet. Future research should determine whether modifying the dose or ratio of supplemental FA in high-starch diets can enhance milk fat synthesis and overall production in early-lactation cows.

As mentioned previously, nutrient delivery to the mammary gland is a key determinant of milk synthesis in dairy cows (Davis and Collier, 1985). The increase in ECM yield observed with starch supplementation was primarily driven by increased yields of milk, fat, and protein, whereas 3.5% FCM improvements in the low-starch and FA-supplemented cows were mainly due to higher milk fat yield. Although we did not determine nutrient digestibility or absorption, the increased milk fat yield observed with FA supplementation was likely driven by greater absorption of mixed FA, particularly C16:0, which plays a central role in milk triglyceride formation due to its ability to enhance the activity of glycerol-3-phosphate acyltransferase. This enzyme has a higher affinity for C16:0 than for other FA, facilitating efficient incorporation of C16:0 into milk fat (Kinsella and Gross, 1973; Tzompa-Sosa et al., 2014). These findings are consistent with previous research evaluating supplemental C16:0-enriched or C16:0 and *cis*-9 C18:1 FA in the immediate postpartum period (de Souza and Lock, 2019; de Souza et al., 2021; Parales-Girón et al., 2025).

With respect to dietary starch, previous studies have shown that inclusion rates of near 30% of DM may reduce milk fat synthesis through alterations in rumen biohydrogenation pathways (Bauman et al., 2011; Boerman et al., 2015b). In our study, dietary starch was 28% DM, and all treatment diets were balanced for fNDF, a critical factor in supporting normal rumen function and minimizing the risk of MFD (NASEM, 2021). This dietary strategy likely contributed to the absence of MFD in our results and the observed increase in milk fat yield, which was likely due to greater DMI in the high-starch diets. This finding aligns with previous studies, which have demonstrated that replacing soyhulls with ground corn while maintaining similar fNDF levels increases milk fat yield (Boerman et al., 2015a). The increase in milk protein observed in our study may be explained by increased DMI, along with enhanced microbial protein synthesis in the rumen, supported by the improved synchronization of ruminal nitrogen and energy supply with starch-rich diets (Oba and Allen, 2003; Rius et al., 2010; Belanche et al.,

2012). This response is consistent with previous studies in early-lactation (Carmo et al., 2015; Daneshvar et al., 2021) and post-peak cows (Akins et al., 2014; Culbertson et al., 2025). Furthermore, ruminal fermentation of starch yields propionate, which could lead to increased hepatic gluconeogenesis and elevated circulating glucose, supporting lactose synthesis and providing carbon skeletons for de novo FA and protein synthesis in the mammary gland (Bionaz and Looor, 2008; Rius et al., 2010; Lin et al., 2016). Even without measurable changes in plasma glucose, the greater propionate yield associated with high-starch diets may enhance milk component yields through increased substrate availability for the mammary gland. This underscores the importance of considering not just total energy, but also carbohydrate source and fermentability when formulating rations. Overall, our results reinforce that careful balancing of dietary starch and fiber is a practical strategy to optimize milk component yields and that supplementation of specific FA may affect these results.

Although treatment diets affected ECM and 3.5% FCM yields, no differences in BW loss were observed. Previous research has shown that increasing dietary starch can reduce BW loss during the first 4 wk of lactation (Andersen et al., 2003) and support subsequent weight gain later in lactation (Piccioli-Cappelli et al., 2014; dos Santos Neto et al., 2025). Feeding glucogenic diets can stimulate the secretion of insulin, a key regulator of lipogenesis in dairy cows (Qiao et al., 2024). Despite this, our results are consistent with previous studies evaluating different levels of dietary starch and sugar in the immediate postpartum period (Penner and Oba, 2009), indicating that dietary starch did not affect plasma insulin levels. This suggests that the additional energy provided by increasing dietary starch was primarily directed toward milk synthesis rather than to body energy reserves. In contrast, the effect of FA supplementation on BW change has been variable in the literature, with C16:0 or C16:0 and *cis*-9 C18:1 supplements reported to reduce, increase, or maintain BW loss (de Souza et al., 2021; de Souza and Lock, 2019; Parales-Girón et al., 2025). In our study, FA supplementation increased plasma NEFA and BHB. Importantly, the increase in plasma NEFA likely reflects increased intestinal absorption of dietary FA rather than enhanced mobilization of adipose tissue (Bionaz et al., 2020). This distinction is crucial because instead of drawing energy from body fat reserves, lactating cows can utilize absorbed dietary FA to support milk production. Such energy partitioning aligns with the genetic prioritization of high-producing cows to prioritize milk synthesis over body energy reserves (Bauman and Currie, 1980; Piantoni and VandeHaar, 2023). These findings highlight the importance of understanding the

distinct effects of dietary starch and FA supplementation for optimizing early lactation nutrition strategies.

Nutritional interventions in the first 3 to 4 weeks of lactation can have persistent effects on the metabolism and production potential of dairy cows (Jørgensen et al., 2016; Rockwell and Allen, 2016; de Souza et al., 2021). Interestingly, we observed a carryover effect of increasing dietary starch, with cows fed the high-starch diet during the treatment period maintaining higher milk and lactose yields after the dietary treatments ceased. This finding suggests that feeding more energy-dense diets rich in starch during the immediate postpartum period may have longer-term benefits for milk synthesis. Ruminal fermentation of starch yields propionate, which serves as the primary gluconeogenic precursor in the liver of dairy cows. Furthermore, glucose has been shown to increase lactose synthesis, cell viability, and proliferation in cultured dairy cow mammary epithelial cells (Lin et al., 2016). In contrast, no carryover effects were observed for FA supplementation. These results contrast with findings from our previous study, in which feeding FA blends with different ratios of C16:0 and *cis*-9 C18:1 in the immediate postpartum period increased ECM yield up to 7 weeks after ceasing the treatment diets (de Souza et al., 2021). A key distinction is that supplemental FA in de Souza et al. (2021) did not decrease DMI as observed in our study, which may account for the differences in the results.

Our study examined the effects of dietary starch level and FA supplementation on milk production of early-lactation cows with treatment diets formulated to have similar CP and fNDF contents. By design, modifying dietary starch content and increasing dietary FA resulted in non-isocaloric diets. We did not collect samples to determine the digestibility of the treatment diets, which would have helped us better explain our results. Because we did not collect digestibility data, it is difficult to determine whether the observed production differences arose from altered rumen fermentation or from postabsorptive metabolism. We considered formulating an isoenergetic control diet using a different fat source; however, different FA profiles have their own metabolic effects, which would confound interpretation (Piantoni et al., 2015; dos Santos Neto et al., 2021; Bales et al., 2024). Although our treatment diets were balanced for fNDF to support rumen function, replacing soyhulls with ground corn could affect rumen fermentability and microbial activity due to differences in nutrient composition and rumen fermentation rates (Ipharraguerre et al., 2002). These differences likely contributed to differences in nutrient availability and microbial activity, potentially affecting our results. Thus, rather than controlling total dietary energy across treatments, we aimed to evaluate how different energy

sources interact to influence animal performance, thereby underlining the importance of considering both ingredient and nutrient selection when designing feeding strategies. The observed interactions between starch and FA likely reflect meaningful differences in how cows utilize and respond to various energy sources, rather than simply being a result of the unequal dietary energy content. Because the diet modifications used in our study rely on standard ingredients and formulations, these are readily applicable to typical production settings and can be used to refine early-lactation feeding recommendations. Continued research (including on-farm trials) is needed to verify these strategies and to monitor potential long-term impacts on cow health and production.

CONCLUSIONS

To our knowledge, no previous studies have evaluated the combined effects of increasing dietary starch and supplementing a FA blend containing 70% C16:0 and 20% *cis*-9 C18:1 during the immediate postpartum period in dairy cows. In this study, HS increased the yields of 3.5% FCM by 4.6 kg/d and milk fat by 0.17 kg/d compared with LS. The LS+FA treatment increased milk fat yield by 0.18 kg/d and tended to increase 3.5% FCM yield by 3.2 kg/d compared with LS. The lack of a synergistic effect likely reflects reduced DMI observed with FA supplementation at 2.0% of diet DM and potential adverse effects on fiber digestibility and rumen biohydrogenation when a high-starch diet and a UFA supplement were combined. Importantly, neither high starch nor FA supplementation exacerbated BW loss. Overall, increasing dietary starch increased milk and milk fat yields, whereas FA supplementation improved milk fat yield only under low-starch conditions.

NOTES

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Nonstandard abbreviations used: CNCPS = Cornell Net Carbohydrate and Protein System; CO = carryover period; FA = fatty acid; fNDF = forage NDF; FR = fresh period; HS = diet containing 28% DM starch and no supplemental FA; HS+FA = diet containing 28% DM starch and a calcium salt containing 70% C16:0 and 20% *cis*-9 C18:1; LS = diet containing 22% DM starch and no supplemental FA; LS+FA = diet containing 22% DM starch and a calcium salt containing 70% C16:0 and 20% *cis*-9 C18:1; MFD = milk fat depression; NEB = negative energy balance; NEFA = nonesterified fatty acids.

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